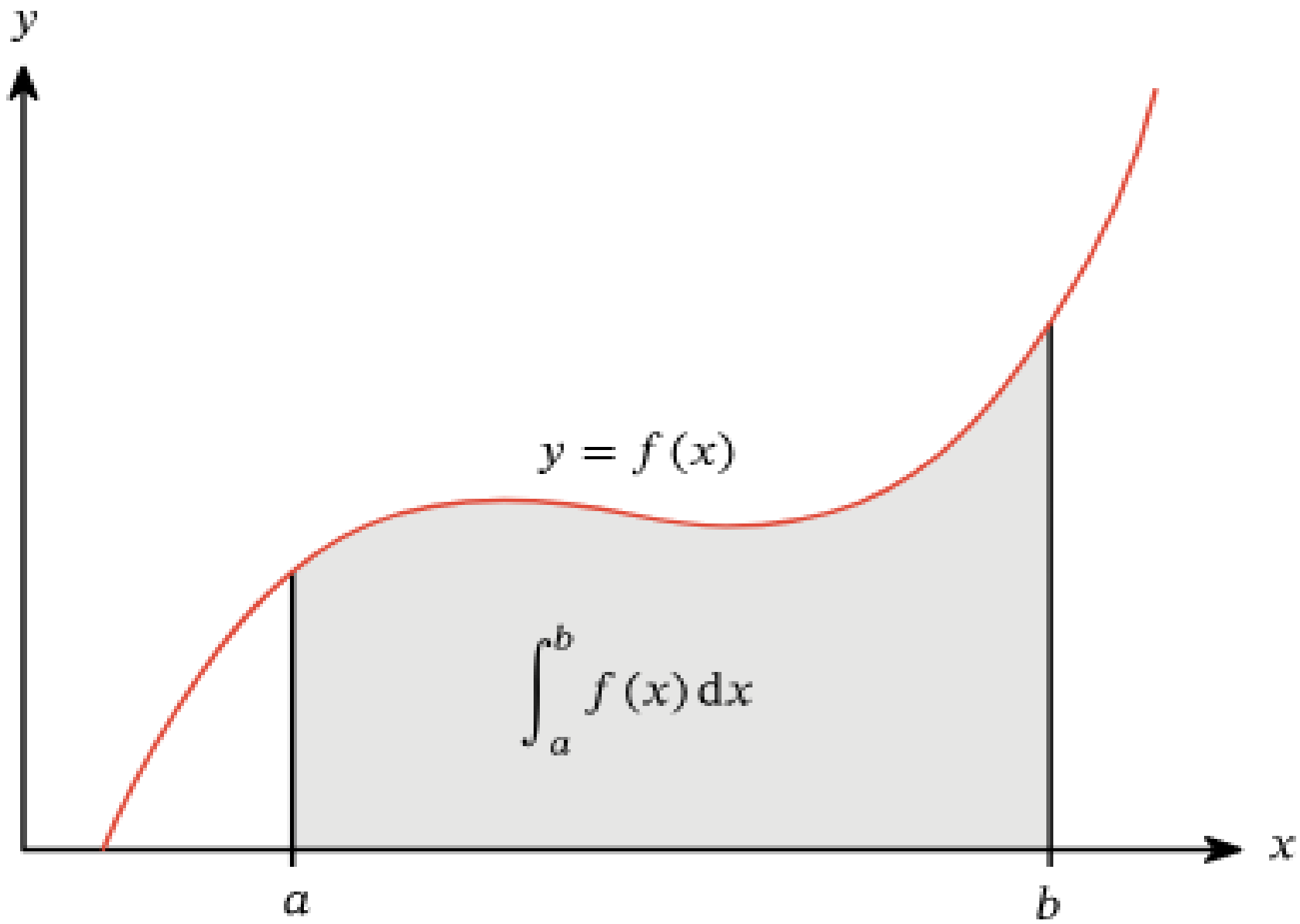
Numerical Integration


y
y = f (x)
∫ₐᵇ f (x) dx
x
a
b

Numerical Integration

- Many real-world integrals cannot be solved analytically.

- Used when:
  - Function is complex or unknown analytically.
  - Data is available only at discrete points.

- Goal: Approximate the area under a curve numerically.

Why We Need Numerical Integration

- Analytical integration often fails for complex or tabulated data.
- Numerical methods use geometric approximations:
  - Rectangles (Midpoint Rule)
  - Trapezoids (Trapezoidal Rule)
  - Parabolas (Simpson’s Rule)
- Enables automation in computer-based solutions.

Midpoint Rule

- Approximates integral using function value at the midpoint.

The *Midpoint Rule* (also called the rectangle rule) approximates the integral by using the value of the function at the midpoint of each subinterval.

Let us approximate the definite integral:

$$I = \int_a^b f(x)\,dx$$

Geometric Approximation (Rectangle Method)

The simplest way to approximate this integral is with the area of a rectangle:

$$\text{Area} = (\text{width}) \times (\text{height})$$

- **Width:** The length of the interval, $h = b - a$.
- **Height:** The value of the function $f(x)$ evaluated at a specific point $\bar{x}$ within $[a, b]$.

The Midpoint Rule specifies that the evaluation point $\bar{x}$ must be the **midpoint** of the interval, $m = \frac{a+b}{2}$.

$$M_1 = f\left(\frac{a+b}{2}\right) \cdot (b - a)$$

$$I \approx \text{Midpoint Rule } (M_1) = (b - a) \cdot f\left(\frac{a+b}{2}\right)$$

Midpoint Rule

We divide the interval $[a, b]$ into n equal parts, each of width:

$$\Delta x = \frac{b-a}{n}$$

For each subinterval $[x_{i-1}, x_i]$, the midpoint is:

$$\bar{x}_i = \frac{x_{i-1} + x_i}{2}$$

The *Midpoint Rule formula* is then given by:

$$I \approx \sum_{i=1}^{n} f(\bar{x}_i)\, \Delta x$$

Derivation of the Midpoint Rule

- Based on Taylor series expansion at midpoint x_m and then applying the integration.

$$f(x) = f(x_m) + f'(x_m)(x - x_m) + (f''(x_m)/2!)(x - x_m)^2 + \ldots$$

- Ignoring higher-order terms gives: $\int_a^b f(x)dx \approx (b - a)\, f(x_m)$

Example Problem

Evaluate $I = \int_0^2 (x^2 + 1)\, dx$ [0,2]

Step 1: a = 0, b = 2

Step 2: Midpoint $x_m = (0 + 2)/2 = 1$

Step 3: $f(1) = 1^2 + 1 = 2$

Step 4: $I \approx (2 - 0)(2) = 4$

Now calculate with 4 intervals

Example Problem

The interval is $[a, b] = [0, 2]$ and the number of intervals is $n = 4$.

$$\Delta x = \frac{b - a}{n} = \frac{2 - 0}{4} = \frac{2}{4} = 0.5$$

The endpoints of the four subintervals are: $[0, 0.5]$, $[0.5, 1]$, $[1, 1.5]$, and $[1.5, 2]$.

Example Problem

$$M_4 = \Delta x \cdot \sum_{i=1}^{4} f(m_i)$$

$$M_4 = 0.5 \cdot (1.0625 + 1.5625 + 2.5625 + 4.0625)$$

$$M_4 = 0.5 \cdot (9.25)$$

$$\mathbf{M_4 = 4.625}$$

Example Problem (Actual)

For comparison, the exact value of the integral is:

$$I = \int_0^2 (x^2 + 1)\, dx = \left[\frac{x^3}{3} + x\right]_0^2$$

$$I = \left(\frac{2^3}{3} + 2\right) - \left(\frac{0^3}{3} + 0\right) = \frac{8}{3} + 2 = \frac{8}{3} + \frac{6}{3} = \frac{14}{3}$$

$$I \approx 4.6666\ldots$$

The Midpoint Rule approximation of 4.625 is very close to the exact value.

- Midpoint Rule is simple and effective.
- Based on midpoint symmetry.
- Advantages:
  - Easy to implement
  - Accurate for smooth functions
- Error -> $(b - a)^3 f''(x_m)$
- Accuracy improves with smaller intervals.

Newton–Cotes Integration

- Family of numerical integration methods.
- Approximates integral using a polynomial through equally spaced points.

General formula:

$I \approx \Sigma\, w_i\, f(x_i)$, where w_i are weights and x_i are nodes.

Newton–Cotes Integration

- Replace $f(x)$ by an approximating polynomial:

$$f_n(x) = a_0 + a_1 x + \cdots + a_n x^n$$

- Integrate the polynomial exactly instead of $f(x)$.

Derivation Overview

We start with the **Lagrange Interpolating Polynomial**:

$$P_n(x) = \sum_{i=0}^{n} f\,(x_i) L_i(x)$$

where each $L_i(x)$is the Lagrange basis polynomial:

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

Then integrate both sides:

$$\int_a^b f\,(x)\,dx \approx \sum_{i=0}^{n} f\,(x_i) \int_a^b L_i\,(x)\,dx$$

The weights $w_i = \int_a^b L_i\,(x)\,dx$ depend only on n and the spacing.

Newton–Cotes Integration

The Newton–Cotes formulas are a **family of integration techniques** that replace $f(x)$with a simpler **polynomial interpolant**, integrate that exactly, and use the result as an approximation.

$$\int_a^b f(x)\,dx \approx \int_a^b P_n(x)\,dx$$

where $P_n(x)$is a polynomial of degree n that passes through $n+1$**equally spaced points**.

After integration, the formula becomes a **weighted sum**:

$$\int_a^b f(x)\,dx \approx \sum_{i=0}^{n} w_i\, f(x_i)$$

where:

x_i :interpolation nodes, $x_i = a + ih$

$$h = \frac{b-a}{n}$$

w_i :precomputed weights (based on polynomial order)

Closed vs. Open Newton–Cotes Formulas

Type	Includes Endpoints?	Typical Use
Closed	Yes — includes (a) and (b)	General smooth functions
Open	No — uses interior points only	Functions undefined at endpoints

Closed Newton–Cotes Methods

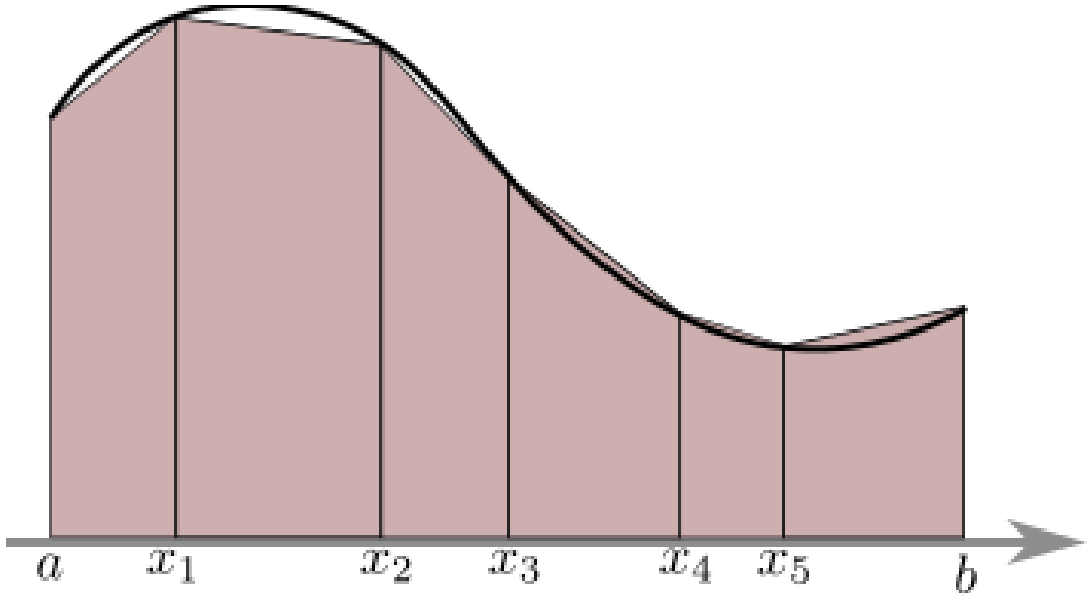
Rule	Polynomial Degree	Formula Type	Name
n = 1	Linear	Closed	**Trapezoidal Rule**
n = 2	Quadratic	Closed	**Simpson's 1/3 Rule**
n = 3	Cubic	Closed	**Simpson's 3/8 Rule**
n = 4	Quartic	Closed	**Boole's Rule**

The Trapezoidal Rule (n=1)

Derivation

Approximate $f(x)$by a straight line joining $f(a)$and $f(b)$:

$$I \approx \frac{b-a}{2}[f(a)+f(b)]$$



The Trapezoidal Rule

Single Subinterval (Trapezoid): Consider a single subinterval $[x_0, x_1]$ of width $h = x_1 - x_0$. The area of the trapezoid formed by the points $(x_0, f(x_0))$ and $(x_1, f(x_1))$ is:

$$A_{\text{trapezoid}} = \text{Average height} \times \text{Width}$$

$$A_1 \approx \frac{f(x_0) + f(x_1)}{2} \cdot h$$

Composite (Multiple-Segment) Rule: Divide the entire interval $[a, b]$ into n equal subintervals, each of width $h = \frac{b-a}{n}$. Let $a = x_0, x_1, x_2, \ldots, x_n = b$ be the endpoints of the subintervals. The total integral is the sum of the areas of the n trapezoids:

$$I \approx \sum_{i=1}^{n} A_i$$

The Trapezoidal Rule

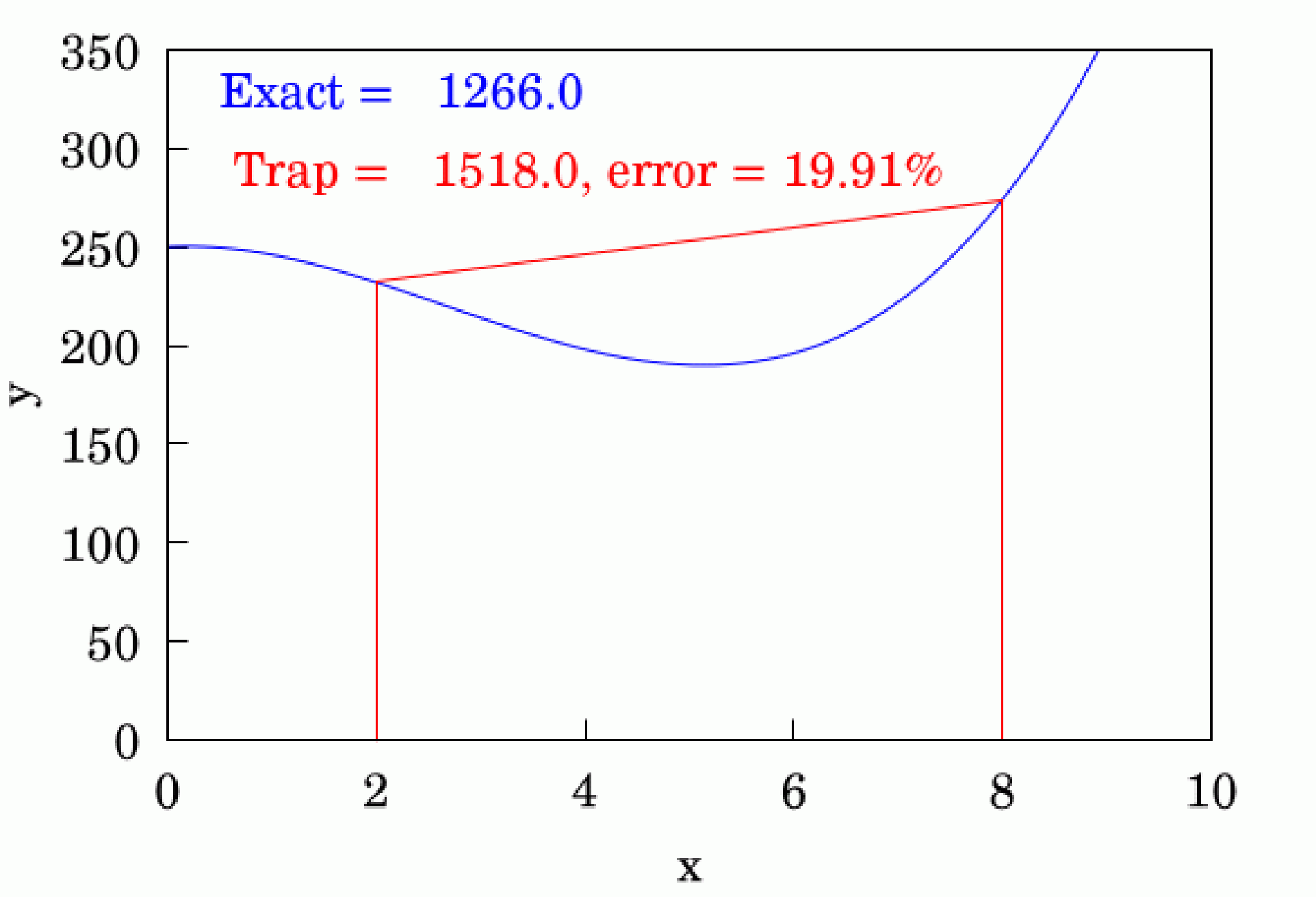
$$I \approx \frac{h}{2}[f(x_0) + f(x_1)] + \frac{h}{2}[f(x_1) + f(x_2)] + \cdots + \frac{h}{2}[f(x_{n-1}) + f(x_n)]$$

Final Formula: Factor out $\frac{h}{2}$ and group the terms:

$$I \approx \frac{h}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)]$$

The **Composite Trapezoidal Rule** formula is:

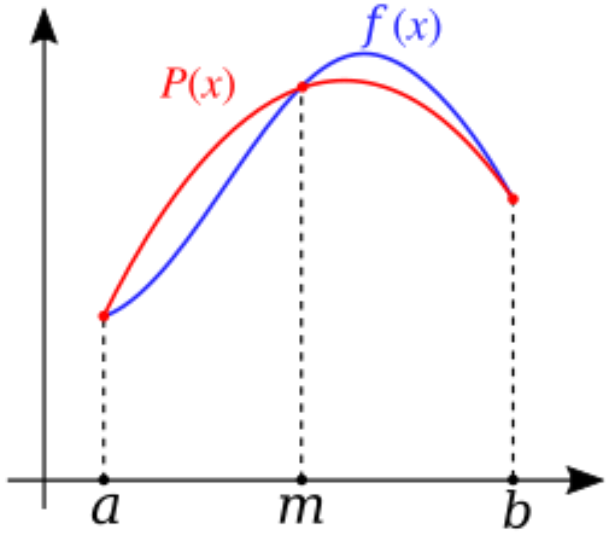
$$I \approx \frac{b-a}{2n}[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)]$$

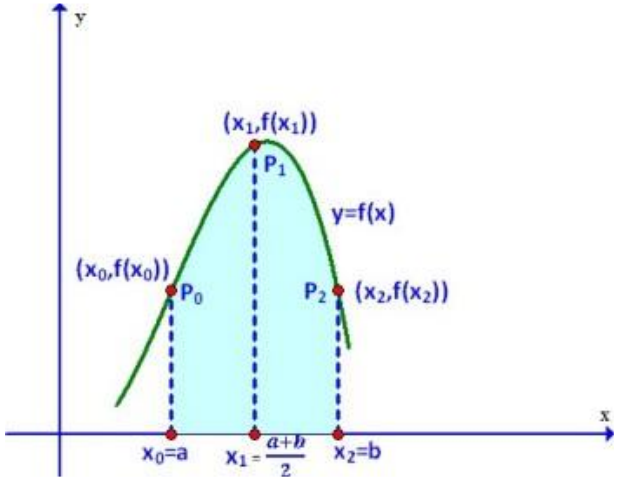

Exact = 1266.0
Trap = 1518.0, error = 19.91%
350
300
250
200
150
100
50
0
y
0
2
4
6
8
10
x

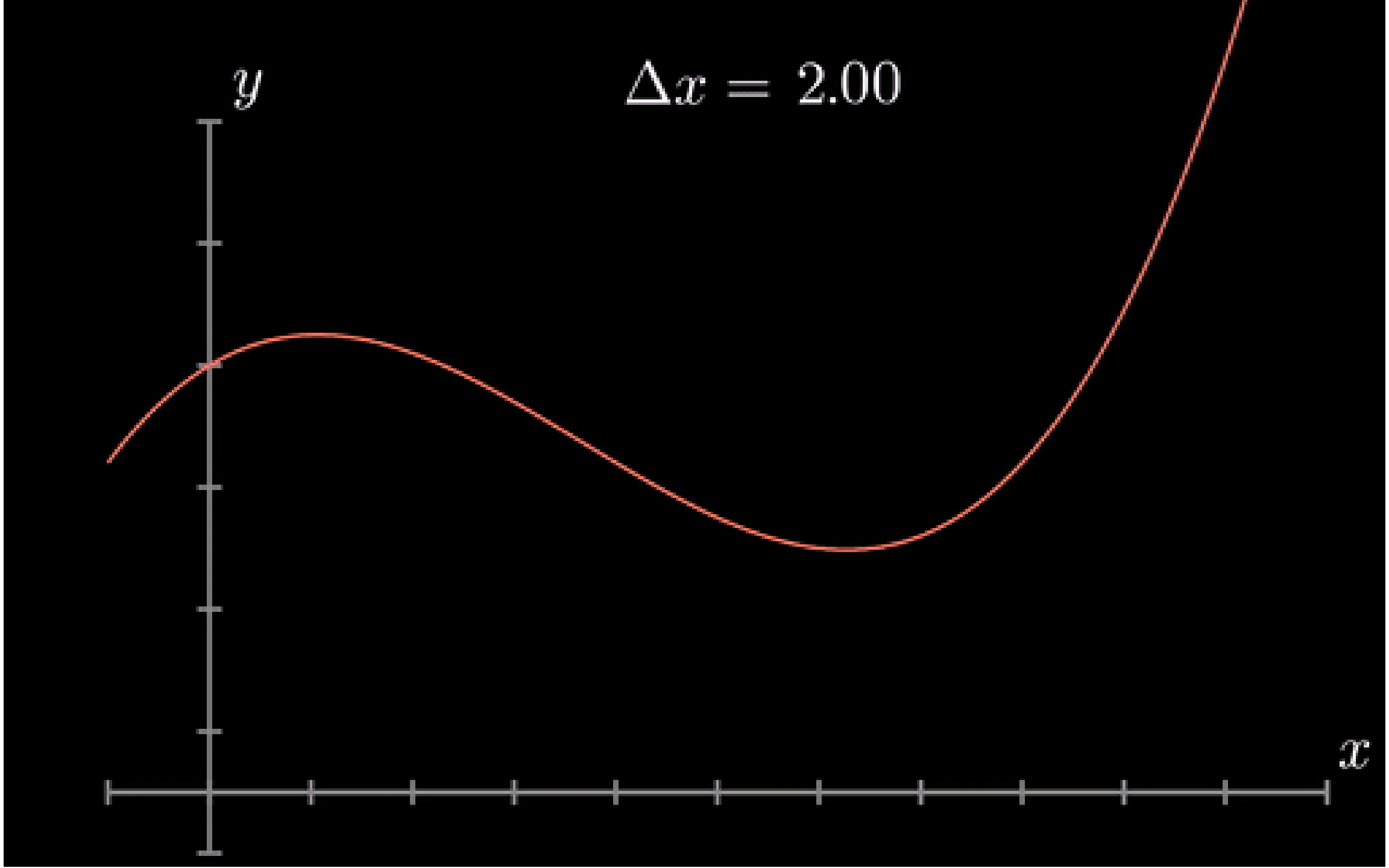
Example

$$I = \int_0^2 (x^2 + 1)\, dx$$

Simpson's 1/3 Rule (n=2)






y
$\Delta x = 2.00$
x

Simpson’s 1/3

Consider a function $f(x)$ defined on $[a, b]$.

Divide it into n equal subintervals of width:

$$h = \frac{b-a}{n}$$

Now take any two subintervals (three points):

$$x_0,\ x_1 = x_0 + h,\ x_2 = x_0 + 2h$$

We approximate $f(x)$ between x_0 and x_2 by a quadratic:

$$f(x) = a + bx + cx^2$$

Simpson’s 1/3

At x_0, x_1, x_2:

$$f(x_0) = a + bx_0 + cx_0^2$$

$$f(x_1) = a + bx_1 + cx_1^2$$

$$f(x_2) = a + bx_2 + cx_2^2$$

We integrate this quadratic from x_0 to x_2:

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right]$$

Simpson’s 1/3

We start with a small portion of the curve between x_0 and x_2, with one midpoint x_1:

$$x_0,\ x_1 = x_0 + h,\ x_2 = x_0 + 2h$$

We’ll approximate $f(x)$ between these three points using a **second-degree polynomial** (a parabola).

Simpson’s 1/3

We assume that $f(x)$ behaves like:

$$f(x) = a + bx + cx^2$$

where a, b, and c are constants that will make the parabola pass through the three known points:

$$(x_0, f_0),\ (x_1, f_1),\ (x_2, f_2)$$

with $f_i = f(x_i)$.

Simpson's 1/3

At each point:

$$\begin{cases} f_0 = a + bx_0 + cx_0^2 \\ f_1 = a + bx_1 + cx_1^2 \\ f_2 = a + bx_2 + cx_2^2 \end{cases}$$

We don't actually need to *solve* for a, b, c — the trick is to find

$$\int_{x_0}^{x_2} f(x)\,dx = \int_{x_0}^{x_2} (a + bx + cx^2)\,dx$$

and then substitute these three equations in such a way that everything simplifies in terms of f_0, f_1, f_2.

Simpson's 1/3

$$I = \int_{x_0}^{x_2} a\,dx + \int_{x_0}^{x_2} bx\,dx + \int_{x_0}^{x_2} cx^2\,dx$$

$$\int_{x_0}^{x_2} f(x)\,dx = a(x_2 - x_0) + \frac{b}{2}(x_2^2 - x_0^2) + \frac{c}{3}(x_2^3 - x_0^3)$$

Since $x_2 - x_0 = 2h$, and $x_1 = x_0 + h$, we can simplify the algebra (after some substitution and grouping) to get a compact relation:

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}\left[f_0 + 4f_1 + f_2\right]$$

That's the **Simpson's 1/3 formula for a single parabolic segment**.

Simpson’s 1/3

We know:

$$\begin{cases} f_0 = a + bx_0 + cx_0^2 \\ f_1 = a + b(x_0 + h) + c(x_0 + h)^2 \\ f_2 = a + b(x_0 + 2h) + c(x_0 + 2h)^2 \end{cases}$$

We’ll use these to eliminate a, b, c.

To make the algebra easier, **shift the coordinate system** so $x_0 = 0$.

Simpson’s 1/3

Then we have:

$$x_1 = h, \quad x_2 = 2h$$

and

$$f_0 = a, \quad f_1 = a + bh + ch^2, \quad f_2 = a + 2bh + 4ch^2$$

Simpson's 1/3

From $f_0 = a$:

$$a = f_0$$

Subtract the first from the second:

$$f_1 - f_0 = bh + ch^2 \quad \Rightarrow \quad b = \frac{f_1 - f_0}{h} - ch$$

Subtract the second from the third:

$$f_2 - f_1 = bh + 3ch^2$$

Substitute b from above:

$$f_2 - f_1 = \left[\frac{f_1 - f_0}{h} - ch\right] h + 3ch^2$$

$$f_2 - f_1 = (f_1 - f_0) + 2ch^2$$

$$c = \frac{f_2 - 2f_1 + f_0}{2h^2}$$

Simpson’s 1/3

$$I = \int_0^{2h} (a + bx + cx^2)\, dx = \left[ax + \tfrac{b}{2}x^2 + \tfrac{c}{3}x^3\right]_0^{2h} = 2ah + 2bh^2 + \tfrac{8}{3}ch^3.$$

We know:

$$a = f_0, \quad b = \frac{f_1 - f_0}{h} - ch, \quad c = \frac{f_2 - 2f_1 + f_0}{2h^2}$$

and from integrating $f(x) = a + bx + cx^2$:

$$I = \int_0^{2h} f(x)\, dx = 2ah + 2bh^2 + \frac{8}{3}ch^3$$

Simpson’s 1/3

$$I = 2(f_0)h + 2h^2\left(\frac{f_1 - f_0}{h} - ch\right) + \frac{8}{3}ch^3$$

Simplify step-by-step:

$$I = 2hf_0 + 2h(f_1 - f_0) - 2ch^3 + \frac{8}{3}ch^3$$

$$I = 2hf_0 + 2hf_1 - 2hf_0 + \left(-2 + \frac{8}{3}\right)ch^3$$

$$I = 2hf_1 + \frac{2}{3}ch^3$$

Now substitute $c = \frac{f_2 - 2f_1 + f_0}{2h^2}$:

$$I = 2hf_1 + \frac{2}{3}h^3 \cdot \frac{f_2 - 2f_1 + f_0}{2h^2}$$

Simplify:

$$I = 2hf_1 + \frac{h}{3}(f_2 - 2f_1 + f_0)$$

$$I = \frac{h}{3}(6f_1 + f_2 - 2f_1 + f_0)$$

$$I = \frac{h}{3}(f_0 + 4f_1 + f_2)$$

Alternate Approach

$$S = \frac{2M + T}{3} = \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

Let $h = \frac{b-a}{2}$ and $m = \frac{a+b}{2}$.

Alternate Approach

$$\boxed{S = \frac{2M + T}{3} = \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]}$$

Let $h = \frac{b-a}{2}$ and $m = \frac{a+b}{2}$.

$$M = (b-a)\,f(m) = 2h\,f(m).$$

$$T = \frac{b-a}{2}\big(f(a) + f(b)\big) = h\big(f(a) + f(b)\big).$$

Alternate Approach

$$\frac{2M+T}{3} = \frac{2(2h\,f(m)) + h\big(f(a)+f(b)\big)}{3} = \frac{h}{3}\Big(4f(m)+f(a)+f(b)\Big).$$

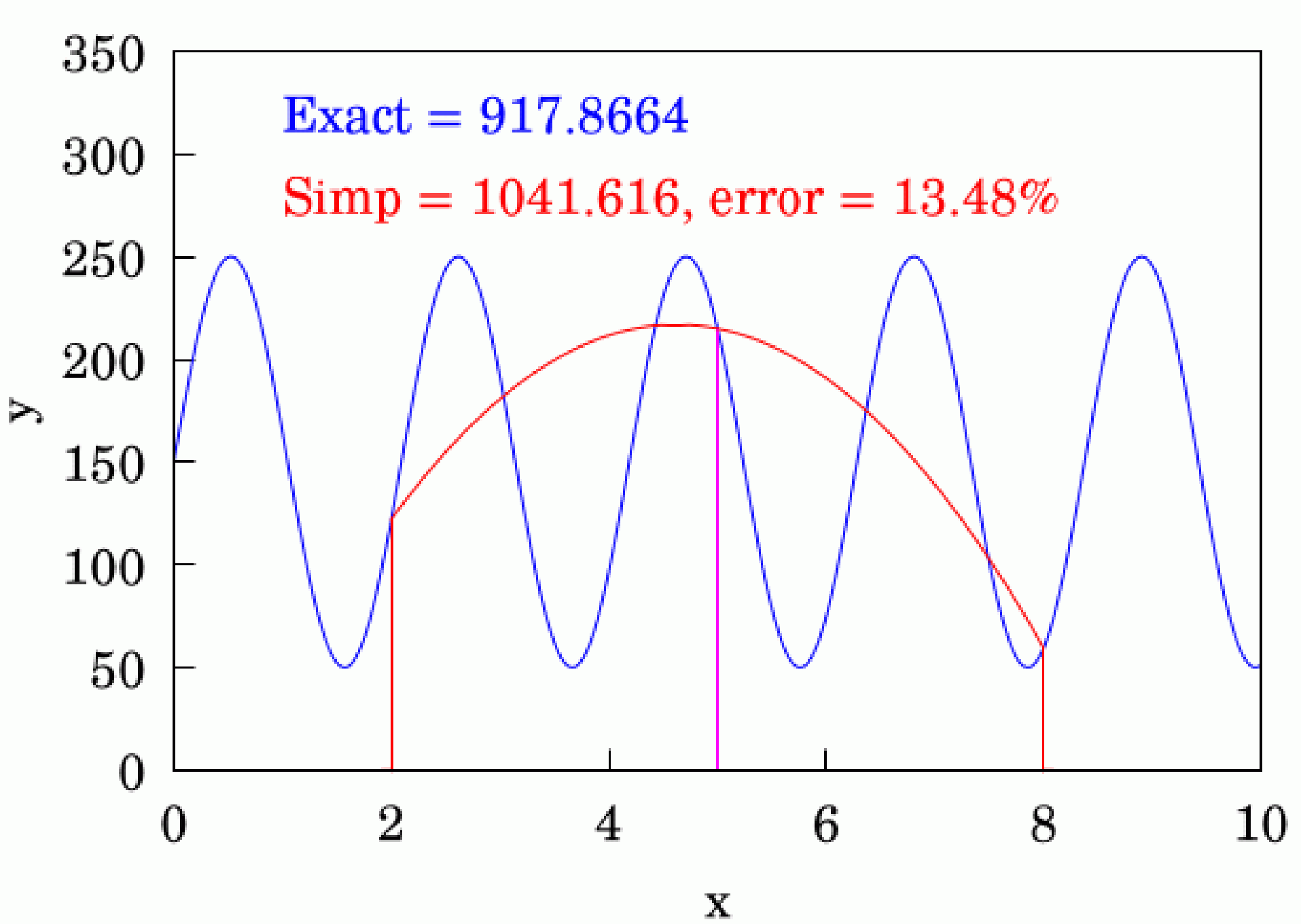
$h = \frac{b-a}{2}$, this becomes

$$\frac{2M+T}{3} = \frac{b-a}{6}\Big(f(a)+4f(m)+f(b)\Big),$$

Weighted Average of Midpoint and Trapezoidal

Simpson’s 1/3 Rule

Rule	Degree (n)	Formula	Order of Error	Exact For
Trapezoidal	1	$\frac{b-a}{2}(f(a)+f(b))$	$O(h^2)$	Linear
Simpson’s 1/3	2	$\frac{b-a}{6}[f(a)+4f(m)+f(b)]$	$O(h^4)$	Quadratic
Simpson’s 3/8	3	$\frac{3h}{8}[f_0+3f_1+3f_2+f_3]$	$O(h^4)$	Cubic
Boole’s	4	Higher-order weights	O(h6)	Quartic


Exact = 917.8664
Simp = 1041.616, error = 13.48%
350
300
250
200
150
100
50
0
y
0
2
4
6
8
10
x

Composite Simpson's 1/3

$$\begin{aligned}\int_a^b f(x)\,dx &\approx \frac{1}{3}h\sum_{i=1}^{n/2}\left[f(x_{2i-2})+4f(x_{2i-1})+f(x_{2i})\right]\\ &= \frac{1}{3}h\left[f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+2f(x_4)+\cdots+2f(x_{n-2})+4f(x_{n-1})+f(x_n)\right]\\ &= \frac{1}{3}h\left[f(x_0)+4\sum_{i=1}^{n/2}f(x_{2i-1})+2\sum_{i=1}^{n/2-1}f(x_{2i})+f(x_n)\right].\end{aligned}$$

Simpson's 3/8 rule

First, consider the definite integral $\int_a^b f(x)dx$.

- We use a cubic polynomial approximation, which requires four points: x_0, x_1, x_2, x_3.
- The interval $[a, b]$ is divided into three equal subintervals.
- Let the step size be $h = \frac{b-a}{3}$.
- The four equally spaced nodes are:
  - $x_0 = a$
  - $x_1 = a + h$
  - $x_2 = a + 2h$
  - $x_3 = a + 3h = b$

Simpson's 3/8 rule

The function $f(x)$ is approximated by a cubic Lagrange interpolating polynomial, $P_3(x)$, that passes through the four points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$, and $(x_3, f(x_3))$

The polynomial is a sum of four terms, each corresponding to a point:

$$P_3(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)$$

where $L_i(x)$ are the Lagrange basis polynomials, defined as:

$$L_i(x) = \prod_{j=0, j\neq i}^{3} \frac{x - x_j}{x_i - x_j}$$

Simpson's 3/8 rule

The definite integral of $f(x)$ is approximated by the definite integral of the Lagrange polynomial:

$$\int_a^b f(x)dx \approx \int_{x_0}^{x_3} P_3(x)dx$$

$$\int_{x_0}^{x_3} P_3(x)dx = \int_{x_0}^{x_3} [L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)]dx$$

$x = a + ht$, $dx = h\,dt$, nodes $t = 0, 1, 2, 3$

Simpson's 3/8 rule

Work in t-space with $x = a + ht$, $dx = h\,dt$, nodes $t = 0, 1, 2, 3$. The Lagrange basis in t is

$$L_0(t) = \frac{(t-1)(t-2)(t-3)}{(0-1)(0-2)(0-3)} = -\frac{(t-1)(t-2)(t-3)}{6},$$

$$L_1(t) = \frac{t(t-2)(t-3)}{(1-0)(1-2)(1-3)} = \frac{t(t-2)(t-3)}{2},$$

$$L_2(t) = \frac{t(t-1)(t-3)}{(2-0)(2-1)(2-3)} = -\frac{t(t-1)(t-3)}{2},$$

$$L_3(t) = \frac{t(t-1)(t-2)}{(3-0)(3-1)(3-2)} = \frac{t(t-1)(t-2)}{6}.$$

Simpson's 3/8 rule

$$\int_0^3 L_0(t)\,dt$$

$$(t-1)(t-2) = t^2 - 3t + 2, \quad (t^2 - 3t + 2)(t-3) = t^3 - 6t^2 + 11t - 6.$$

So

$$L_0(t) = -\frac{t^3 - 6t^2 + 11t - 6}{6} = \frac{-t^3 + 6t^2 - 11t + 6}{6}.$$

$$\int_0^3 L_0(t)\,dt = \left[-\frac{t^4}{24} + \frac{t^3}{3} - \frac{11t^2}{12} + t\right]_0^3 = -\frac{81}{24} + 9 - \frac{99}{12} + 3 = \frac{3}{8}.$$

Simpson's 3/8 rule

$$\int_0^3 L_1(t)\,dt$$

$$t(t-2)(t-3) = t(t^2-5t+6) = t^3-5t^2+6t, \quad L_1(t) = \frac{t^3-5t^2+6t}{2}.$$

Integrate:

$$\int_0^3 L_1(t)\,dt = \frac{1}{2}\left[\frac{t^4}{4} - \frac{5t^3}{3} + 3t^2\right]_0^3 = \frac{1}{2}\left(\frac{81}{4} - 45 + 27\right) = \frac{9}{8}.$$

Simpson's 3/8 rule

$$\int_0^3 L_2(t)\,dt$$

$$t(t-1)(t-3) = t(t^2-4t+3) = t^3-4t^2+3t, \quad L_2(t) = -\frac{t^3-4t^2+3t}{2}.$$

Integrate:

$$\int_0^3 L_2(t)\,dt = -\frac{1}{2}\left[\frac{t^4}{4} - \frac{4t^3}{3} + \frac{3t^2}{2}\right]_0^3 = -\frac{1}{2}\left(\frac{81}{4} - 36 + \frac{27}{2}\right) = \frac{9}{8}.$$

Simpson's 3/8 rule

$$\int_0^3 L_3(t)\,dt$$

$$t(t-1)(t-2) = t(t^2 - 3t + 2) = t^3 - 3t^2 + 2t, \quad L_3(t) = \frac{t^3 - 3t^2 + 2t}{6}.$$

Integrate:

$$\int_0^3 L_3(t)\,dt = \frac{1}{6}\left[\frac{t^4}{4} - t^3 + t^2\right]_0^3 = \frac{1}{6}\left(\frac{81}{4} - 27 + 9\right) = \frac{3}{8}.$$

Simpson's 3/8 rule

$$\int_0^3 L_0\,dt = \frac{3}{8}, \quad \int_0^3 L_1\,dt = \frac{9}{8}, \quad \int_0^3 L_2\,dt = \frac{9}{8}, \quad \int_0^3 L_3\,dt = \frac{3}{8}.$$

$$\int_{x_0}^{x_3} [L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)]dx$$

Composite Simpson's 3/8 rule

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f_0 + 3f_1 + 3f_2 + f_3\right]$$

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f_0 + 3\sum_{\substack{1\le i\le n-1\\ i\not\equiv 0 \pmod 3}} f_i + 2\sum_{\substack{3\le i\le n-3\\ i\equiv 0 \pmod 3}} f_i + f_n\right], \quad n = 3m.$$

Composite Simpson's 3/8 rule - Explanation

we have $n = 3m$, $h = (b-a)/n$, nodes $x_i = a + ih$, and for each block of three subintervals:

$$\int_{x_{3k}}^{x_{3k+3}} f(x)\,dx \;\approx\; \frac{3h}{8}\Big[f_{3k} + 3f_{3k+1} + 3f_{3k+2} + f_{3k+3}\Big], \quad k = 0, 1, \ldots, m-1.$$

$$\sum_{k=0}^{m-1} \Big[f_{3k} + 3f_{3k+1} + 3f_{3k+2} + f_{3k+3}\Big]$$

Composite Simpson's 3/8 rule - Explanation

- **k = 0** contributes

$$f_0 + 3f_1 + 3f_2 + f_3.$$

- **k = 1** contributes

$$f_3 + 3f_4 + 3f_5 + f_6.$$

- **k = 2** contributes

$$f_6 + 3f_7 + 3f_8 + f_9.$$

:eep going ... up to

- **k = m-2** contributes

$$f_{3m-6} + 3f_{3m-5} + 3f_{3m-4} + f_{3m-3}.$$

- **k = m-1** contributes

$$f_{3m-3} + 3f_{3m-2} + 3f_{3m-1} + f_{3m}.$$

Composite Simpson's 3/8 rule

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f_0 + 3\sum_{\substack{1\le i\le n-1\\ i\not\equiv 0 \pmod 3}} f_i + 2\sum_{\substack{3\le i\le n-3\\ i\equiv 0 \pmod 3}} f_i + f_n\right], \quad n = 3m.$$

Gaussian Quadrature

Gaussian Quadrature

In numerical integration, **Gaussian Quadrature** is a technique that approximates an integral using *optimally chosen points* (called **nodes**) and *weights*.

It aims to compute

$$I = \int_a^b f(x)\,dx$$

with very high accuracy, even for a small number of sample points.

$$I = \int_a^b f(x)\,dx = \int_{-1}^{1} f(t)\,dt = \sum_{i=1}^{n} w_i f(t_i)$$

$$x = c_1 t + c_2$$

When

$x = a,\ t = -1$

$x = b,\ t = 1$

$$a = -c_1 + c_2$$

$$b = c_1 + c_2$$

$$c_2 = \frac{a+b}{2}, \quad c_1 = \frac{b-a}{2}$$

$$x = \frac{(b-a)}{2}t + \frac{(a+b)}{2}$$

$$\frac{dx}{dt} = \frac{b-a}{2} \quad \text{so} \quad dx = \frac{(b-a)}{2}dt$$

$$I = \int_a^b f(x)\,dx = \int_{-1}^{1} f\left(\frac{b-a}{2}t + \frac{a+b}{2}\right) \cdot \frac{(b-a)}{2}\,dt$$

$$f(\ t\ ) = f\left(\frac{b-a}{2}t + \frac{a+b}{2}\right)$$

$$I = \int_a^b f(x)\,dx = \int_{-1}^{1} f(t) \cdot \frac{(b-a)}{2}\,dt = \frac{b-a}{2}\int_{-1}^{1} f(t)\,dt$$

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2}\sum_{i=1}^{N} w_i f\left(\frac{b-a}{2}t_i + \frac{a+b}{2}\right)$$

Number of points, n	Points, x_i		Weights, w_i	
1	0		2	
2	$\pm\frac{1}{\sqrt{3}}$	±0.57735...	1	
3	0		$\frac{8}{9}$	0.888889...
	$\pm\sqrt{\frac{3}{5}}$	±0.774597...	$\frac{5}{9}$	0.555556...
4	$\pm\sqrt{\frac{3}{7}-\frac{2}{7}\sqrt{\frac{6}{5}}}$	±0.339981...	$\frac{18+\sqrt{30}}{36}$	0.652145...
	$\pm\sqrt{\frac{3}{7}+\frac{2}{7}\sqrt{\frac{6}{5}}}$	±0.861136...	$\frac{18-\sqrt{30}}{36}$	0.347855...
5	0		$\frac{128}{225}$	0.568889...
	$\pm\frac{1}{3}\sqrt{5-2\sqrt{\frac{10}{7}}}$	±0.538469...	$\frac{322+13\sqrt{70}}{900}$	0.478629...
	$\pm\frac{1}{3}\sqrt{5+2\sqrt{\frac{10}{7}}}$	±0.90618...	$\frac{322-13\sqrt{70}}{900}$	0.236927...

Example

$$I = \int_0^1 e^{-x^2}\, dx$$

$$x = \frac{b-a}{2}\,t + \frac{a+b}{2} = \frac{1}{2}t + \frac{1}{2}, \quad dx = \frac{1}{2}dt,$$

so the quadrature on $[0, 1]$ becomes

$$\int_0^1 f(x)\,dx \;\approx\; \frac{1}{2}\sum_{i=1}^{n} w_i\, f\left(\tfrac{1}{2}t_i + \tfrac{1}{2}\right).$$

Example

$$I = \int_0^1 e^{-x^2}\,dx$$

$n = 1$

Nodes/weights: $t_1 = 0,\ w_1 = 2.$

$$I \approx \tfrac{1}{2}\left[2\, f\left(\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}\right)\right] = f(0.5) = e^{-0.25} \approx \mathbf{0.7788007831}.$$

$n = 2$

Nodes/weights: $t_{1,2} = \pm\frac{1}{\sqrt{3}},\ w_1 = w_2 = 1.$

Mapped points:

$$x_{1,2} = \tfrac{1}{2}\left(\tfrac{a+b}{1} \pm \tfrac{1}{\sqrt{3}}\right) = \tfrac{1}{2}\left(1 \pm \tfrac{1}{\sqrt{3}}\right) \approx 0.2113248654,\ 0.7886751346.$$

$$I \approx \tfrac{1}{2}\Big[f(x_1) + f(x_2)\Big] = \tfrac{1}{2}\Big(e^{-x_1^2} + e^{-x_2^2}\Big) \approx \mathbf{0.7465946883}.$$